

Lab 13

Submitted by : inshal nasir

Section : A

Roll no: 38

Submitted to : sir shoaib

Task 0

```
ATI@DESKTOP-KQE4P80 MINGW64 ~/Lab13 (main)
$ gh codespace ssh
? Choose codespace: 23-22412-038-sys/Lab13 [main]: vigilant halibut
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-1030-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.
```

Task 1

```
@23-22412-038-sys →/workspaces/Lab13 (main) $ mkdir -p ~/Lab13
cd ~/Lab13
```

```
@23-22412-038-sys →~/Lab13 $ touch main.tf
ls -la
total 12
drwxr-xr-x 2 codespace codespace 4096 Jan  2 10:08 .
drwxr-x--- 1 codespace codespace 4096 Jan  2 10:07 ..
-rw-r--r-- 1 codespace codespace    0 Jan  2 10:08 main.tf
```

```
GNU nano 7.2 main.tf *
provider "aws" {
  shared_config_files      = ["~/.aws/config"]
  shared_credentials_files = ["~/.aws/credentials"]
}
```

```
@23-22412-038-sys →~/Lab13 $ aws sts get-caller-identity
{
  "UserId": "623705167686",
  "Account": "623705167686",
  "Arn": "arn:aws:iam::623705167686:root"
}
```

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see any changes that are required for your infrastructure. All Terraform commands should now work.

```
@23-22412-038-sys →~/Lab13 $ terraform apply -auto-approve
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

+ create

Terraform will perform the following actions:

```
# aws_iam_group.developers will be created
```

```
@23-22412-038-sys →~/Lab13 $ terraform output
```

```
group_details = {
  "group_arn" = "arn:aws:iam::623705167686:group/groups/developers"
  "group_name" = "developers"
  "unique_id" = "AGPAZCN5ZS5DPX3WSBTHJ"
}
```

```
@23-22412-038-sys →~/Lab13 $ █
```

User groups (1) [Info](#)

A user group is a collection of IAM users. Use groups to specify permissions

☐	Group name	▼	Users
☐	developers		

Summary [Edit](#)

User group name developers	Creation time January 02, 2026, 02:21 (UTC-08:00)	ARN arn:aws:iam::623705167686:group/groups/developers
-------------------------------	--	--

[Users](#) [Permissions](#) [Access Advisor](#)

Users in this group (0) [Back to console](#) [Add users](#)

Task 2

```
# IAM user: loadbalancer
resource "aws_iam_user" "lb" {
  name      = "loadbalancer"
  path      = "/users/"
}
```

Apply complete! Resources: 2 added, 0 changed, 0 destroyed.

Outputs:

```
group_details = {
  "group_arn" = "arn:aws:iam::623705167686:group/groups/developers"
}
```

Outputs:

```
group_details = {
    "group_arn" = "arn:aws:iam::623705167686:group/groups/developers"
    "group_name" = "developers"
    "unique_id" = "AGPAZCN5ZS5DPX3WSBTHJ"
}
```

```
@23-22412-038-sys → ~/Lab13 $ terraform output
```

```
group_details = {
    "group_arn" = "arn:aws:iam::623705167686:group/groups/developers"
    "group_name" = "developers"
    "unique_id" = "AGPAZCN5ZS5DPX3WSBTHJ"
}
```

```
user_details = {
    "unique_id" = "AIDAZCN5ZS5DOWWNH4ETE"
    "user_arn" = "arn:aws:iam::623705167686:user/users/loadbalancer"
    "user_name" = "loadbalancer"
}
```

loadbalancer [Info](#)

Summary

ARN

 arn:aws:iam::623705167686:user/users/loadbalancer

Created

January 02, 2026, 02:28 (UTC-08:00)

Permissions

Groups (1)

Tags (1)

Security

User groups membership

A user group is a collection of IAM users. Use groups to specify

☐ | **Group name**

☐ [developers](#)

Task 3

```
# Attach AmazonEC2FullAccess to developers group
resource "aws_iam_group_policy_attachment" "developer_ec2_fullaccess" {
  group      = aws_iam_group.developers.name
  policy_arn = "arn:aws:iam::aws:policy/AmazonEC2FullAccess"
}
```

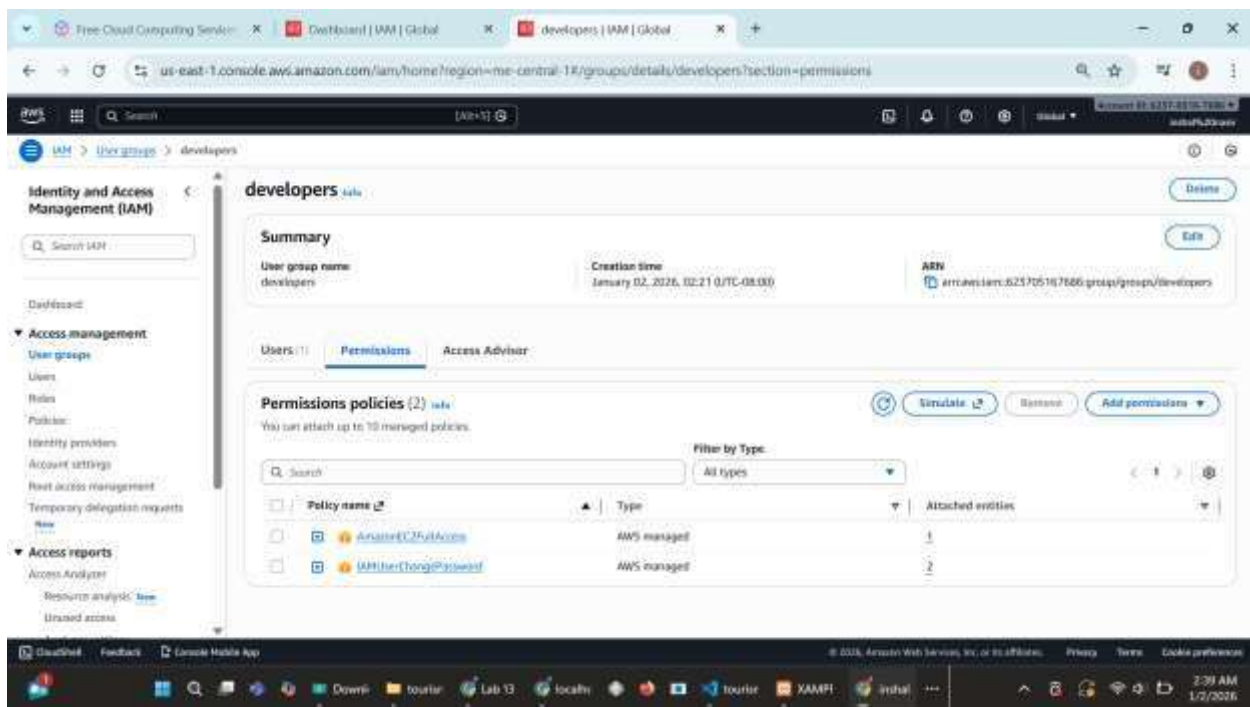
     

Terraform will perform the following actions:

```
# aws_iam_user.lb will be created
+ resource "aws_iam_user" "lb" {
  + arn              = (known after apply)
  + force_destroy    = true
```

@23-22412-038-sys → ~/Lab13 \$ terraform apply -auto-approve

```
group_details = {
  "group_arn" = "arn:aws:iam::623705167686:group/groups/developers"
  "group_name" = "developers"
  "unique_id" = "AGPAZCN5ZS5DPX3WSBTHJ"
}
user_details = {
  "unique_id" = "AIDAZCN5ZS5DOWWNH4ETE"
```



Task 4

```

GNU nano 7.2                                variables.tf *
variable "iam_password" {
  description = "Temporary password for the IAM user"
  type        = string
  sensitive    = true
  default      = "1dontKnow"
}

```

```

GNU nano 7.2                                create-login-profile.sh *
#!/usr/bin/env bash
set -euo pipefail

USERNAME="$1"
PASSWORD="$2"

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
nano create_login_profile.sh
@23-22412-038-sys →~/Lab13 $ chmod +x create-login-profile.sh
@23-22412-038-sys →~/Lab13 $

```

```

GNU nano 7.2                                main.tf *
resource "null_resource" "create_login_profile" {
  triggers = {
    password_hash = sha256(var.iam_password)
    user          = aws_iam_user.lb.name
  }
}

```

```

@23-22412-038-sys →~/Lab13 $ terraform output
group_details = {
  "group_arn" = "arn:aws:iam::623705167686:group/groups/developers"
  "group_name" = "developers"
  "unique_id" = "AGPAZCN5ZS5DPX3WSBTHJ"
}

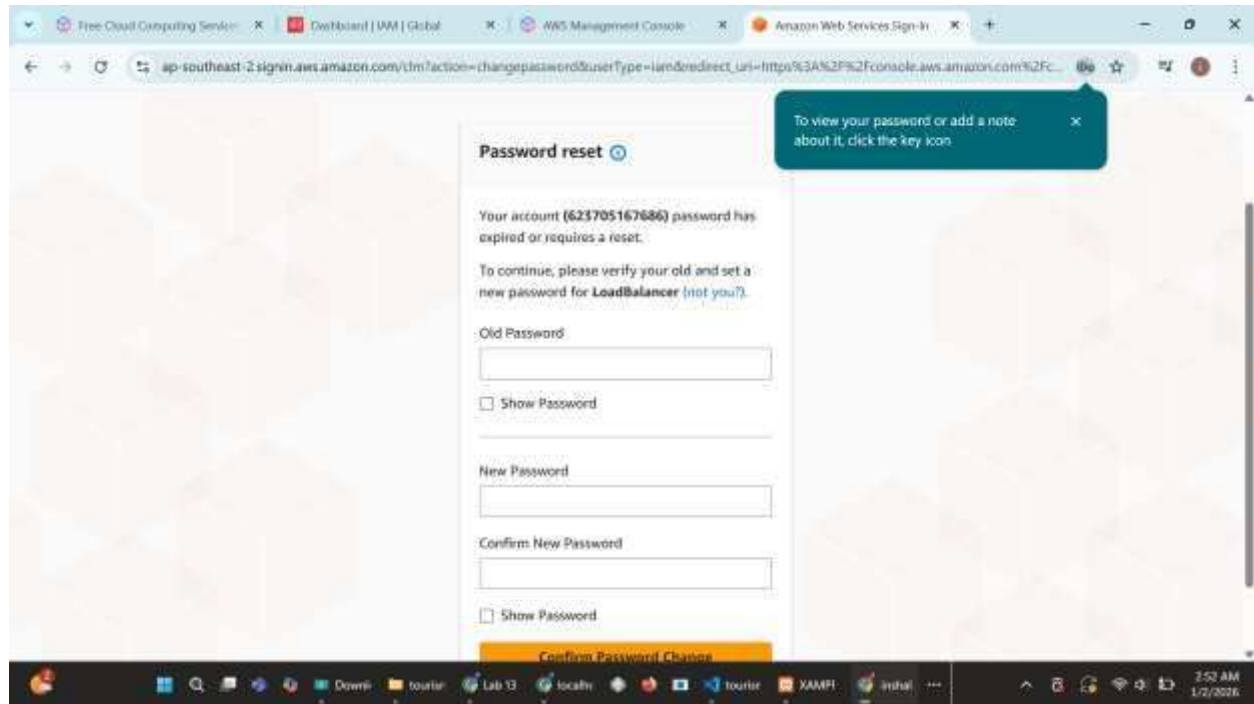
}

user_details = {
  "unique_id" = "AIDAZCN5ZS5DOWWNH4ETE"
  "user_arn" = "arn:aws:iam::623705167686:user/users/loadbalancer"
  "user_name" = "loadbalancer"
}

```



```
@23-22412-038-sys →~/Lab13 $ aws iam get-login-profile --user-name loadbalancer
{
  "LoginProfile": {
    "UserName": "loadbalancer",
    "CreateDate": "2026-01-02T10:49:10+00:00",
    "PasswordResetRequired": true
  }
}
```



Task 5

```
GNU nano 7.2 main.tf
# IAM Access Key for loadbalancer
resource "aws_iam_access_key" "lb_access_key" {
  user = aws_iam_user.lb.name
}

+ access_key_secret = (sensitive value)
aws_iam_access_key.lb_access_key: Creating...
aws_iam_access_key.lb_access_key: Creation complete after 1s [id=AKIAZCN5ZS5DFB02LNNN]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.
```

```

access_key_id = "AKIAZCN5ZS5DFBO2LNNN"
access_key_secret = <sensitive>
group_details = {
    "group_arn" = "arn:aws:iam::623705167686:group/groups/developers"
    "group_name" = "developers"
    "unique_id" = "AGPAZCN5ZS5DPX3WSBTHJ"
}

```

loadbalancer [Info](#) [Delete](#)

Summary

ARN arn:aws:iam::623705167686:user/users/loadbalancer	Console access Enabled without MFA	Access key 1 AKIAZCN5ZS5DFBO2LNNN - Active ⓘ Never used. Created today.
Created January 02, 2026, 02:28 (UTC-08:00)	Last console sign-in Today	Access key 2 Create access key

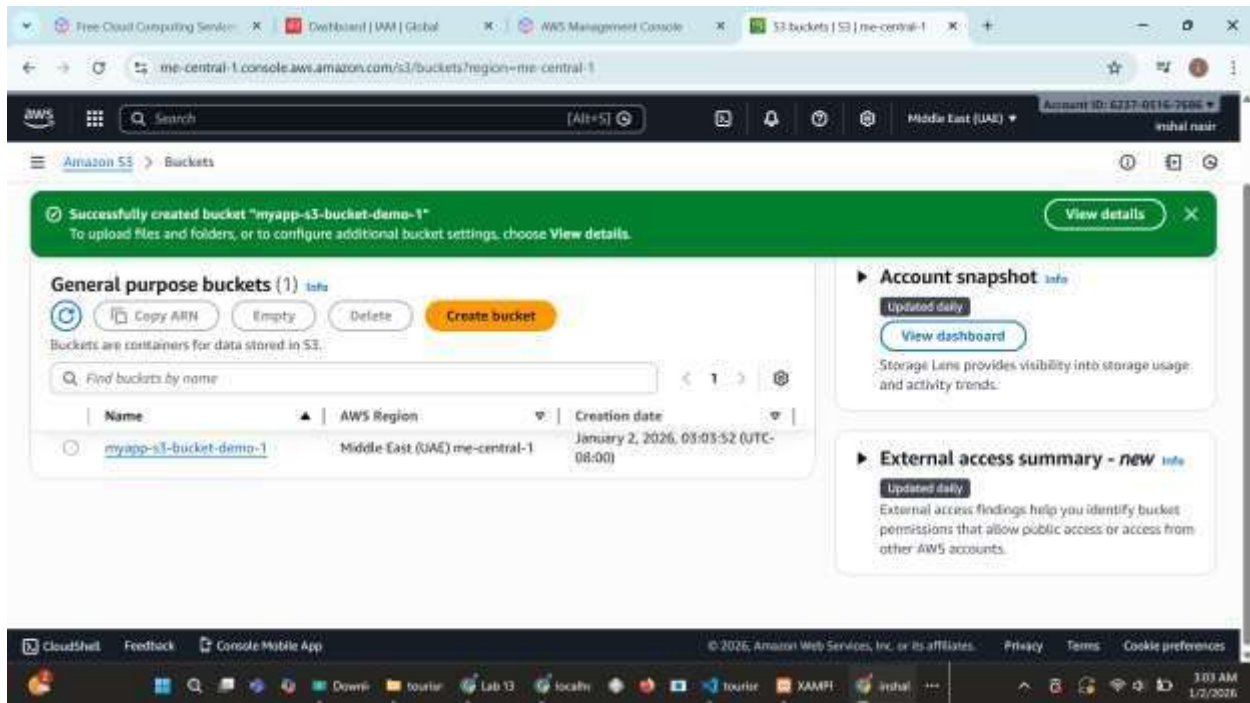
[Permissions](#) | [Groups \(1\)](#) | [Tags \(1\)](#) | [Security credentials](#) | [Last Accessed](#)

Console sign-in [Manage console access](#)

Console sign-in link https://623705167686.signin.aws.amazon.com/console	Console password Updated 10 minutes ago (2026-01-02 02:49 PST)
Last console sign-in 7 minutes ago (2026-01-02 02:52 PST)	

Task 6

The screenshot shows the AWS Management Console interface for the 'me-central-1' region. The main content area is titled 'General purpose buckets (0)' and includes a search bar and a 'Create bucket' button. The left sidebar contains a navigation menu with categories like 'Buckets', 'Access management and security', and 'Storage management and insights'. The top navigation bar displays the user's account information, including the name 'initial.nair' and the region 'Middle East (UAE)'.



```
@23-22412-038-sys →~/Lab13 $ terraform init -migrate-state
```

- Using previously-installed hashicorp/aws v5.100.0
- Using previously-installed hashicorp/null v3.2.4

Terraform has been successfully initialized!

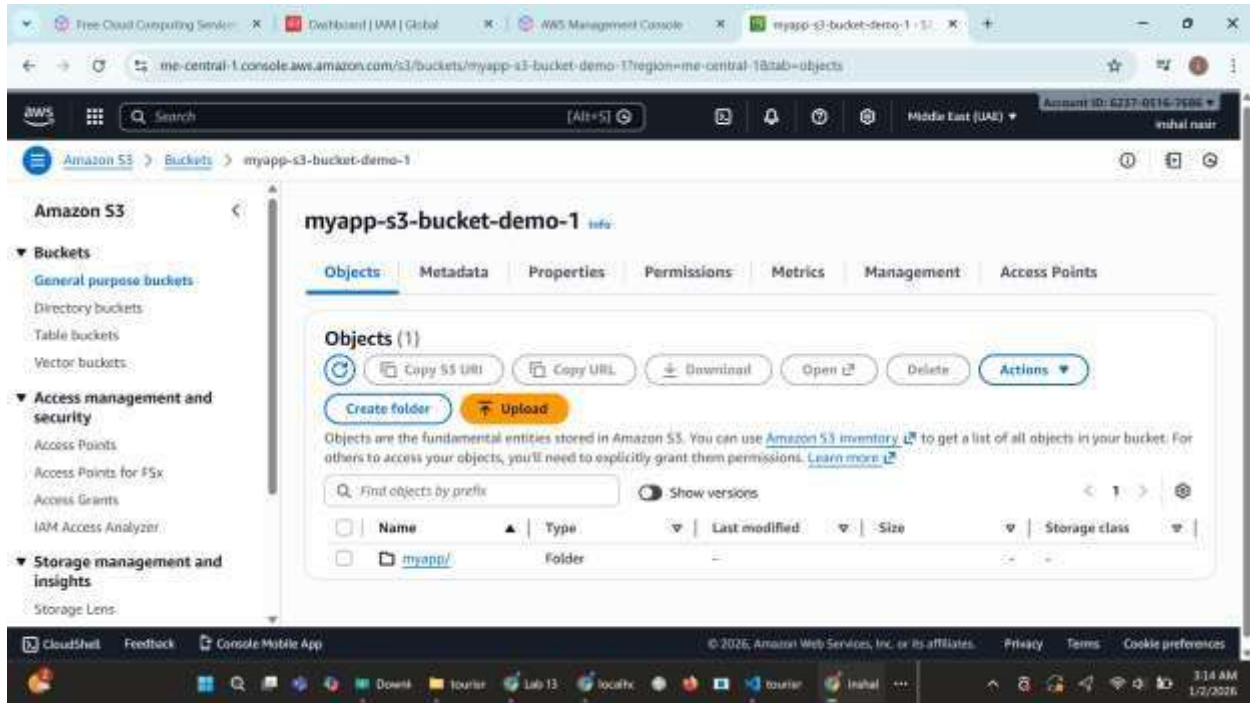
You may now begin working with Terraform. Try running "terraform plan" to see any changes that are required for your infrastructure. All Terraform commands should now work.

```
@23-22412-038-sys →~/Lab13 $ terraform apply -auto-approve -var="iam_password=MySecurePass123!"
```

No changes. Your infrastructure matches the configuration.

Terraform has compared your real infrastructure against your configuration and found no differences, so no changes are needed.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

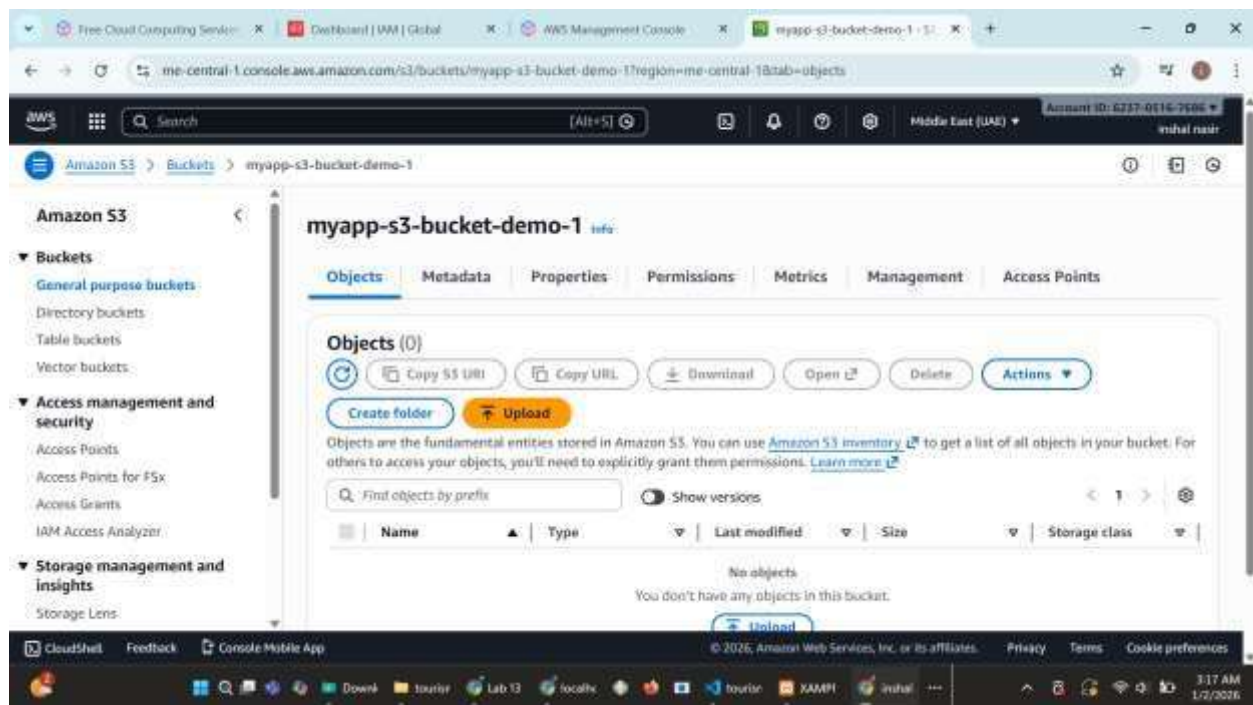


```
@23-22412-038-sys →~/Lab13 $ ls -la terraform.tfstate*  
-rw-r--r-- 1 codespace codespace 0 Jan 2 11:10 terraform.tfstate  
-rw-r--r-- 1 codespace codespace 6882 Jan 2 11:10 terraform.tfstate.backup  
@23-22412-038-sys →~/Lab13 $
```

```
@23-22412-038-sys →~/Lab13 $ terraform destroy -auto-approve  
aws_iam_group_policy_attachment.developer_ec2_fullaccess: Destruction complete af  
ter 1s  
aws_iam_group.developers: Destroying... [id=developers]  
aws_iam_group.developers: Destruction complete after 0s  
aws_iam_user.lb: Destruction complete after 2s
```

Destroy complete! Resources: 7 destroyed.

```
@23-22412-038-sys →~/Lab13 $
```

Task 7

```

GNU nano 7.2                                users.csv *
user_name
Michael
Dwight
Jim
Pam

GNU nano 7.2                                locals.tf *
locals {
  users = csvdecode(file("users.csv"))
}

```

```
@23-22412-038-sys →~/Lab13 $ terraform init -upgrade
```

- Finding hashicorp/null versions matching "~> 3.2"...
- Finding hashicorp/aws versions matching "~> 5.0"...
- Using previously-installed hashicorp/aws v5.100.0
- Using previously-installed hashicorp/null v3.2.4

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see

PROBLEMS OUTPUT TERMINAL ... bash - Lab13

```
@23-22412-038-sys →~/Lab13 $ terraform apply -auto-approve -var="iam_password=MySecurePass123!"
```

```
    "access_key_id" = "AKIAZCN5ZS5DNSPIMN6L"
    "user_arn"      = "arn:aws:iam::623705167686:user/users/Kevin"
    "user_unique_id" = "AIDAZCN5ZS5DJNJFIRG7D"
  }
  "Meredith" = {
    "access_key_id" = "AKIAZCN5ZS5DE7U7CVIU"
    "user_arn"      = "arn:aws:iam::623705167686:user/users/Meredith"
```

```
@23-22412-038-sys →~/Lab13 $ terraform output
```

```
all_access_key_secrets = <sensitive>
```

```
all_users_details = {
```

```
  "Andy" = {
    "access_key_id" = "AKIAZCN5ZS5DDMCDTPPG"
    "user_arn"      = "arn:aws:iam::623705167686:user/users/Andy"
    "user_unique_id" = "AIDAZCN5ZS5DDUMIDKGQ3"
  }
```

```
  }
  "Angela" = {
    "access_key_id" = "AKIAZCN5ZS5DED2XGOJO"
    "user_arn"      = "arn:aws:iam::623705167686:user/users/Angela"
    "user_unique_id" = "AIDAZCN5ZS5DL6WDEJWEV"
  }
}
```



```
}
"Charles" = {
  "access_key_id" = "AKIAZCN5ZS5DLQRVIKWQ"
  "user_arn" = "arn:aws:iam::623705167686:user/users/Charles"
  "user_unique_id" = "AIDAZCN5ZS5DGN3KWMLKM"
}
"Clark" = {
}
"Creed" = {
  "access_key_id" = "AKIAZCN5ZS5DEYU5LIIA"
  "user_arn" = "arn:aws:iam::623705167686:user/users/Creed"
  "user_unique_id" = "AIDAZCN5ZS5DGG4SUB7XF"
}
"Darryl" = {
  "access_key_id" = "AKIAZCN5ZS5DLQ2GVWMQ"
  "user_arn" = "arn:aws:iam::623705167686:user/users/Darryl"
  "user_unique_id" = "AIDAZCN5ZS5DBVOPF5RE7"
}
"David" = {
  "access_key_id" = "AKIAZCN5ZS5DC2XVMR32"
  "user_arn" = "arn:aws:iam::623705167686:user/users/David"
  "user_unique_id" = "AIDAZCN5ZS5DEI5V24FT6"
}
"Dwight" = {
  "access_key_id" = "AKIAZCN5ZS5DEPJLNIWD"
  "user_arn" = "arn:aws:iam::623705167686:user/users/Dwight"
  "user_unique_id" = "AIDAZCN5ZS5DKSBBTURE3"
}
"Erin" = {
  "access_key_id" = "AKIAZCN5ZS5DAY3J74W5"
  "user_arn" = "arn:aws:iam::623705167686:user/users/Erin"
  "user_unique_id" = "AIDAZCN5ZS5DDEZR7EQUU"
}
"Gabe" = {
```

```
,
"Gabe" = {
  "access_key_id" = "AKIAZCN5ZS5DOLTKDXDG"
  "user_arn" = "arn:aws:iam::623705167686:user/users/Gabe"
  "user_unique_id" = "AIDAZCN5ZS5DCAKPPGRHA"
}

"Holly" = {
  "access_key_id" = "AKIAZCN5ZS5DPH6QDGGGL"
  "user_arn" = "arn:aws:iam::623705167686:user/users/Holly"
  "user_unique_id" = "AIDAZCN5ZS5DKZLEZQ5B7"
}

}

"Jan" = {
  "access_key_id" = "AKIAZCN5ZS5DE4W367OM"
  "user_arn" = "arn:aws:iam::623705167686:user/users/Jan"
  "user_unique_id" = "AIDAZCN5ZS5DJBXI40AT"
}

}

"Jim" = {
  "access_key_id" = "AKIAZCN5ZS5DNIEWRXZT"
  "user_arn" = "arn:aws:iam::623705167686:user/users/Jim"
  "user_unique_id" = "AIDAZCN5ZS5DHAA4CKHCC"
}

"Jo" = {
  "access_key_id" = "AKIAZCN5ZS5DI2004YNY"
  "user_arn" = "arn:aws:iam::623705167686:user/users/Jo"
}

}

"Kelly" = {
  "access_key_id" = "AKIAZCN5ZS5DLES5QLTE"
  "user_arn" = "arn:aws:iam::623705167686:user/users/Kelly"
  "user_unique_id" = "AIDAZCN5ZS5DEO6HDRJLB"
}

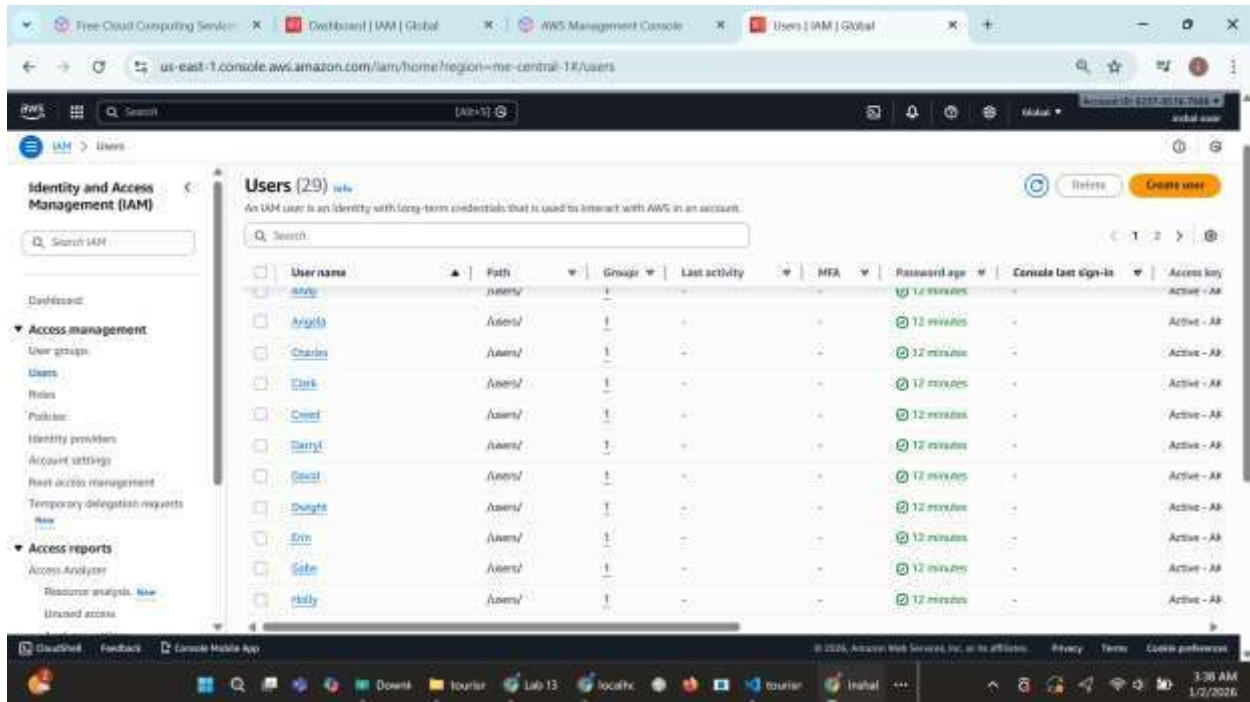
}

"Kevin" = {
```

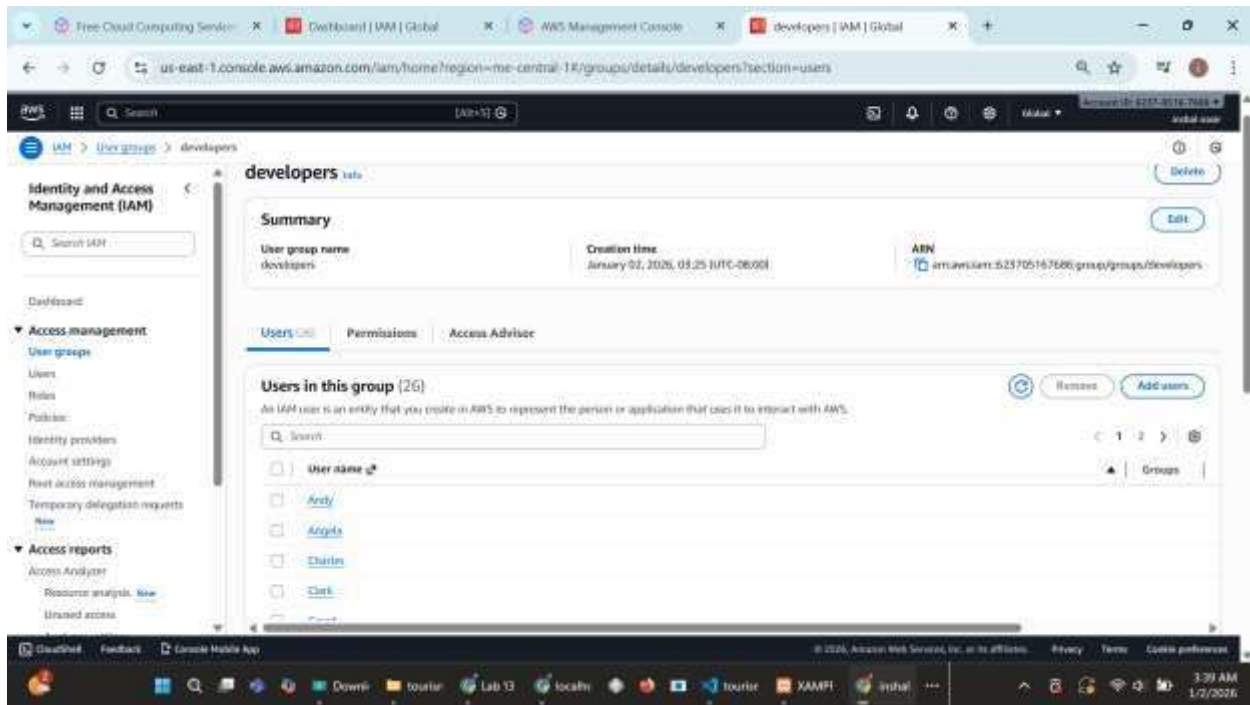
```

"Oscar" = {
  "access_key_id" = "AKIAZCN5ZS5DPQUA2PH3"
  "user_arn" = "arn:aws:iam::623705167686:user/users/Oscar"
  "user_unique_id" = "AIDAZCN5ZS5DGRXJH4ZNR"
}
"Pam" = {
  "access_key_id" = "AKIAZCN5ZS5DOGHOFBIH"
  "user_arn" = "arn:aws:iam::623705167686:user/users/Pam"
}
"Robert" = {
  "access_key_id" = "AKIAZCN5ZS5DNYDZS5VH"
  "user_arn" = "arn:aws:iam::623705167686:user/users/Robert"
  "user_unique_id" = "AIDAZCN5ZS5DMKLKZTY6A"
}

```



Gabe	/users/	1	-	-	12 minutes	-	Active - AR
Holly	/users/	1	-	-	12 minutes	-	Active - AR
Jan	/users/	1	-	-	12 minutes	-	Active - AR
Jim	/users/	1	-	-	12 minutes	-	Active - AR
Jo	/users/	1	-	-	12 minutes	-	Active - AR
Kelly	/users/	1	-	-	8 minutes	-	Active - AR
Kevin	/users/	1	-	-	12 minutes	-	Active - AR
Meredith	/users/	1	-	-	12 minutes	-	Active - AR
Michael	/users/	1	-	-	12 minutes	-	Active - AR



☐	Charles	
☐	Clark	
☐	Creed	
☐	Darryl	
☐	David	
☐	Dwight	
☐	Erin	
☐	Gabe	
☐	Holly	
☐	Jan	
☐	Jim	
☐	Jo	
☐	Kelly	
☐	Kevin	
☐	Kelly	
☐	Kevin	
☐	Meredith	
☐	Michael	
☐	Oscar	1
☐	Pam	1

Pam Info Delete

Summary

ARN <code>arn:aws:iam::625705167686:user/users/Pam</code>	Console access Enabled without MFA	Access key 1 AKIAZCN5ZSSDOGHOFBIH - Active Never used. Created today.
Created January 02, 2026, 03:25 (UTC-08:00)	Last console sign-in Never	Access key 2 Create access key

Permissions | Groups (1) | Tags (2) | **Security credentials** | Last Accessed

Console sign-in Manage console access

Console sign-in link: `https://console.aws.amazon.com/console/home?region=us-east-1#/iam/users/Pam` | Console password: `*****`

Free Cloud Computing | Docker | IAM | Global | AWS Management Console | developers | IAM | Global | myapp-s3-bucket-demo

me-central-1.console.aws.amazon.com/s3/buckets/myapp-s3-bucket-demo-1?prefix=myapp/&2Fregion=me-central-1&tab=objects

Amazon S3 | Buckets | myapp-s3-bucket-demo-1 | myapp/

Amazon S3

- Buckets
 - General purpose buckets
 - Directory buckets
 - Table buckets
 - Vector buckets
- Access management and security
 - Access Points
 - Access Points for FSx
 - Access Grants
 - IAM Access Analyzer
- Storage management and insights
 - Storage Lens

myapp/ Copy S3 URI

Objects | Properties

Objects (1)

[Copy S3 URI](#) [Copy URL](#) [Download](#) [Open](#) [Delete](#) [Actions](#)

[Create folder](#) [Upload](#)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 Inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix: ☐ Show versions

☐	Name	Type	Last modified	Size	Storage class
☐	terraform.tfstate	tfstate	January 2, 2026, 03:30:18 (UTC-08:00)	95.0 KB	Standard

CloudShell | Feedback | Console Mobile App

© 2026, Amazon Web Services, Inc. or its affiliates. | Privacy | Terms | Cookie preferences

3:42 AM 1/2/2026

Cleanup

```
aws_iam_user.users["Darryl"]: Destruction complete after 2s
aws_iam_user.users["Michael"]: Destruction complete after 2s
aws_iam_user.users["Clark"]: Destruction complete after 2s
aws_iam_user.users["Stanley"]: Destruction complete after 2s
aws_iam_user.users["Jim"]: Destruction complete after 5s
aws_iam_user.users["Peter"]: Destruction complete after 8s
```

Destroy complete! Resources: 107 destroyed.

Free Cloud Computing | Dashboard | IAM | Global | AWS Management Console | developers | IAM | Global | Users | IAM | Global

us-east-1.console.aws.amazon.com/iam/home?region=us-east-1#/users

Identity and Access Management (IAM)

Search IAM

Dashboard

Access management

- User groups
- Users**
- Roles
- Policies
- Identity providers
- Account settings
- Root access management
- Temporary delegation requests

Access reports

- Access Analyzer
- Resource analyzer
- Unshared access

Users (0)

An IAM user is an identity with long-term credentials that is used to interact with AWS in an account.

Search

Username	Path	Groups	Last activity	MFA	Password age	Console last sign-in	Access key ID
No resources to display							

3:46 AM 1/2/2026

Free Cloud Computing | Dashboard | IAM | Global | AWS Management Console | developers | IAM | Global | User groups | IAM | Global

us-east-1.console.aws.amazon.com/iam/home?region=us-east-1#/groups

Identity and Access Management (IAM)

Search IAM

Dashboard

Access management

- User groups**
- Users
- Roles
- Policies
- Identity providers
- Account settings
- Root access management
- Temporary delegation requests

Access reports

- Access Analyzer
- Resource analyzer
- Unshared access

User groups (0)

A user group is a collection of IAM users. Use groups to specify permissions for a collection of users.

Search

Group name	Users	Permissions	Creation time
No resources to display			

3:47 AM 1/2/2026

